

ON THE JOINT DISTRIBUTIONS OF THE LARGEST AND SMALLEST LATENT ROOTS OF TWO RANDOM MATRICES (NONCENTRAL CASE)

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1. Introduction

The joint distributions of the largest and smallest roots of certain random matrices are useful in the application of some test procedures in multivariate analysis. In the null case, Sugiyama (1970) derived the joint density of the largest and smallest roots of the central Wishart matrix. Waikar (1971) derived an alternate expression (which is better from the point of view of computing than that given by Sugiyama) and also derived the joint density of the largest and smallest roots of the MANOVA matrix in the null case. In the present paper, the author has derived the exact expressions for the joint nonnull density of the largest and smallest roots of (i) $S_1 S_2^{-1}$ matrix, where S_1 and S_2 are independently distributed Wishart matrices and (ii) Wishart matrix. The expressions for the above densities are in terms of zonal polynomials.

2. Preliminaries

If A is a square, nonsingular matrix its inverse and determinant are denoted respectively by A^{-1} and $|A|$. The transpose, trace and exponential of the trace of a matrix B are denoted respectively by B' , $\text{tr } B$ and $\text{etr } B$. Also I_p and O_p denote respectively a $p \times p$ identity matrix and a $p \times p$ null matrix. In addition, we define as in James (1964)

$$p^F_{q(a_1, \dots, a_p; b_1, \dots, b_q; S)} = \sum_{k=0}^{\infty} \sum_{\kappa} \frac{(a_1)_{\kappa} \dots (a_p)_{\kappa}}{(b_1)_{\kappa} \dots (b_q)_{\kappa}} \frac{C_{\kappa}(S)}{k!}$$

and

$$p^F_{q(a_1, \dots, a_p; b_1, \dots, b_q; S, T)} = \sum_{k=0}^{\infty} \sum_{\kappa} \frac{(a_1)_{\kappa} \dots (a_p)_{\kappa}}{(b_1)_{\kappa} \dots (b_q)_{\kappa}} \frac{C_{\kappa}(S) C_{\kappa}(T)}{C_{\kappa}(I_p) k!}$$

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where S and T are $p \times p$ symmetric matrices and $\kappa = (k_1, \dots, k_p)$ is a partition of the integer k satisfying (i) $k_1 \geq k_2 \geq \dots \geq k_p \geq 0$ and (ii) $k_1 + \dots + k_p = k$. Further

$$(a)_\kappa = \prod_{i=1}^p (a - (i-1)/2)_{k_i}$$

$$(a)_k = a(a+1)\dots(a+1-k)$$

and finally $C_\kappa(S)$ is the zonal polynomial as defined in James (1964) and satisfies $(\text{tr } S)^k = \sum_{\kappa} C_\kappa(S)$. A special case of the above is

$${}_1F_0(a; S) = |I_p - S|^{-a}.$$

Remark 1:

Note that if one of the a_i 's above is a negative integer say $a_1 = -n$ then for $k \geq pn + 1$ all the coefficients vanish so that the function ${}_pF_q$ reduces to a (finite) polynomial of degree pn (see Constantine (1963), p. 1276). Further, throughout the paper, whenever a partition say $\kappa = (k_1, \dots, k_p)$ of a nonnegative integer k is defined, it will be implied that

- (i) $k_1 \geq \dots \geq k_p \geq 0$ and
- (ii) $k_1 + \dots + k_p = k$

The following three lemmas are needed in the sequel.

Lemma 2.1:

Let k and d be two nonnegative integers and let $\kappa = (k_1, \dots, k_p)$ and $\delta = (d_1, \dots, d_p)$ denote partitions respectively of k and d . Further let $G = \text{diag}(g_1, \dots, g_p)$. Then

$$(2.1) \quad C_\kappa(G) \cdot C_\delta(G) = \sum_{\beta} g_{\kappa, \delta}^{\beta} C_{\beta}(G)$$

where $\beta = (b_1, \dots, b_p)$ is a partition of the integer $k + d = b$.

Lemma 2.2:

Let G be as defined in Lemma 2.1 and further let $G_1 = \text{diag}(1, G)$. Let $\kappa = (k_1, \dots, k_{p+1})$ be a partition of a nonnegative integer k . Then

$$(2.2) \quad C_\kappa(G_1) = \sum_{t=0}^k \sum_{\tau} b_{\kappa, \tau} C_{\tau}(G)$$

where $\tau = (t_1, \dots, t_p)$ is a partition of t .

The above two lemmas are stated in Khatri and Pillai (1968). The g -coefficients in

(2.1) and the b-coefficients in (2.2) were tabulated by Khatri and Pillai (1968) for various values of the arguments and can be obtained from them.

Throughout this paper, the following notations will be used:

$$\Gamma_p(a) = \pi^{p(p-1)/4} \prod_{i=1}^p \Gamma(a - (i-1)/2)$$

$$\prod_{i < j=t}^p (a_i - a_j) = \prod_{i=t}^{p-1} \prod_{j=i+1}^p (a_i - a_j), 0 \leq t \leq p.$$

The following lemma can be proved by making trivial modification in the proof of the Lemma given in Sugiyama (1967).

Lemma 2.3:

Let $R = \text{diag}(r_1, \dots, r_{p-1})$ where $0 < r_1 < \dots < r_{p-1} < 1$ and let $R_1 = \text{diag}(r_1, \dots, r_{p-1}, 1)$. Further let $\kappa = (k_1, \dots, k_p)$ be a partition of the positive integer k . Then

$$(2.3) \quad D_{\kappa}^p(t) = \int_{0 < r_1 < \dots < r_{p-1} < 1} |R|^{t-(p+1)/2} |I_{p-1}-R| C_{\kappa}(R_1) \prod_{i=1}^{p-1} (r_i - r_j) dR$$

$$= (pt+k)/(\Gamma_p(p/2) / \pi^{p^2/2}) C_{\kappa}(I_p / (\Gamma_p(t, \kappa) \Gamma_p((p+1)/2)) / \Gamma_p(t+(p+1)/2, \kappa))$$

where

$$\Gamma_p(a, \kappa) = \pi^{p(p-1)/4} \prod_{i=1}^p \Gamma(a + k_i - (i-1)/2).$$

3. Joint density of largest and smallest roots of $S_1 S_2^{-1}$

Let X be a $p \times n$ matrix whose columns are distributed independently as multivariate normal with covariance matrix Σ_1 and means given by $E(X) = 0$ and let Y be a $p \times m$ matrix whose column vectors are distributed independently as multivariate normal with covariance matrix Σ_2 and $E(Y) = 0$. Let $S_1 = XX'$ and $S_2 = YY'$ be independently distributed and let $p \leq (m, n)$. Thus S_1 and S_2 have independent Wishart distributions. In addition, let $\lambda_1 \geq \dots \geq \lambda_p > 0$ be the latent roots of $\Sigma_1 \Sigma_2^{-1}$ and $\ell_1 > \dots > \ell_p > 0$ be the latent roots of $S_1 S_2^{-1}$. The joint density of ℓ_1 and ℓ_p is derived below.

The joint density of $\ell_1, \dots, \ell_p$ given by Constantine (see James (1964)) is

$$(3.1) \quad h_1(\ell_1, \dots, \ell_p) = c(p, n, m) |\Omega|^{-\frac{1}{2}n} {}_1F_0\left(\frac{1}{2}(n+m); -\Omega^{-1}, L\right) |L|^{\frac{1}{2}(n-p-1)} \prod_{i < j=1}^p (\ell_i - \ell_j),$$

$$\infty > \ell_1 > \dots > \ell_p > 0$$

where $\Omega = \text{diag}(\lambda_1, \dots, \lambda_p)$, $L = \text{diag}(\ell_1, \dots, \ell_p)$ and

$$c(p, n, m) = \pi^{p^2/2} \Gamma_p(\frac{1}{2}(n+m)) / (\Gamma_p(p/2) \Gamma_p(n/2) \Gamma_p(m/2)).$$

In this section we restrict our discussion to those values of Ω for which the power series expansion of ${}_1F_0$ in (3.1) above is convergent.

Now, on making the transformation $\ell_1 = \ell_1$, $f_i = 1 - \ell_i/\ell_1$ $i=2, \dots, p$ in (3.1), we obtain the joint density of $\ell_1, f_2, \dots, f_p$ as

$$(3.2) \quad h_2(\ell_1, f_2, \dots, f_p) = c(p, n, m) |\Omega|^{-\frac{1}{2}n} \ell_1^{\frac{1}{2}np-1} |F| |I_{p-1}-F|^{-\frac{1}{2}(n-p-1)} \\ \cdot \prod_{i>j=2}^p (f_i - f_j) \sum_{k=0}^{\infty} \sum_{\kappa} \frac{((n+m)/2)_{\kappa} \ell_1^k C_{\kappa}(-\Omega^{-1})}{k! C_{\kappa}(I_p)} \\ \cdot \sum_{t=0}^k \sum_{\tau} b_{\kappa, \tau} C_{\tau}(F)$$

$$0 < \ell_1 < \infty, 0 < f_2 < \dots < f_p < 1$$

where $F = \text{diag}(f_2, \dots, f_p)$, $\kappa = (k_1, \dots, k_p)$ and $\tau = (t_1, \dots, t_{p-1})$ are partitions of the integers k and t respectively and the constants $b_{\kappa, \tau}$ are given by Lemma 2.2. Now, on making the transformation $\ell_1 = \ell_1$, $r_i = f_i/f_p$, $i=2, \dots, p-1$, $f_p = f_p$ and then integrating out $r_2, \dots, r_{p-1}$ over the surface $0 < r_2 < \dots < r_{p-1} < 1$ (using Lemma 2.3) we get the joint density of ℓ_1 and f_p from which we obtain the joint density of ℓ_1 and ℓ_p as

$$(3.3) \quad h_3(\ell_1, \ell_p) = c(p, n, m) |\Omega|^{-\frac{1}{2}n} \ell_1^{\frac{1}{2}p(n-p-1)} (\ell_1 - \ell_p)^{\frac{1}{2}p(p+1)-2} \\ \cdot \sum_{k=0}^{\infty} \sum_{\kappa} \frac{((n+m)/2)_{\kappa} C_{\kappa}(-\Omega^{-1})}{k! C_{\kappa}(I_p)} \sum_{t=0}^k \sum_{\tau} b_{\kappa, \tau} \\ \cdot \sum_{d=0}^{\infty} \sum_{\delta} \frac{(-(n-p-1)/2)_{\delta}}{d!} \ell_1^{k-t-d} (\ell_1 - \ell_p)^{t+d} \sum_{\beta} g_{\tau, \delta}^{\beta} D_{\beta}^{p-1} \left(\frac{p+2}{2} \right)$$

$$\infty > \ell_1 > \ell_p > 0$$

where $\kappa, \tau, b_{\kappa, \tau}$ are as defined above for (3.2), $\delta = (d_1, \dots, d_{p-1})$ is a partition of the integer d , $\beta = (b_1, \dots, b_{p-1})$ is a partition of the integer $t+d$ and the constants $g_{\tau, \delta}^{\beta}$ and

$D_{\beta}^{p-1}((p+2)/2)$ are given by Lemmas 2.1 and 2.3 respectively.

Remark 2:

An important observation is that in the special case when $(n-p-1)/2$ is an integer the summation over d becomes finite (see Remark 1) in (3.3) which means the nonnull density of ℓ_1 and ℓ_p involves only one infinite sum.

The joint nonnull density of ℓ_1 and ℓ_p derived above is useful to compute the power of Roy's test (see Roy (1957)) for the equality of the covariance matrices of two multivariate normal distributions.

4. Joint Density of the largest and smallest roots of Wishart matrix

Let X be a $p \times n$ matrix whose columns are distributed independently as multivariate normal with zero mean vector and covariance matrix Σ and let $p \leq n$. Further, let $S = XX'$ and let $\ell_1 > \ell_2 > \dots > \ell_p > 0$ be the characteristic roots of S . Thus S has a Wishart distribution and the joint density of its roots $\ell_1, \dots, \ell_p$ as derived by James (1964) is

$$(4.1) \quad h_4(\ell_1, \dots, \ell_p) = k(p, n) |\Sigma|^{-\frac{1}{2}n} {}_0F_0 \left(-\frac{1}{2}\Sigma^{-1}, L \right) |L|^{\frac{1}{2}(n-p-1)} \prod_{i=1}^p (\ell_i - \ell_j),$$

$$\infty > \ell_1 > \dots > \ell_p > 0$$

where $L = \text{diag}(\ell_1, \dots, \ell_p)$ and $k(p, n) = \pi^{p^2/2} / (2^{pn/2} \Gamma_p(n/2) \Gamma_p(p/2))$.

On using the same procedure as in the previous section, we obtain the joint density of ℓ_1 and ℓ_p as

$$(4.2) \quad h_5(\ell_1, \ell_p) = k(p, n) |\Sigma|^{-\frac{1}{2}n} \ell_1^{\frac{1}{2}p(n-p-1)} (\ell_1 - \ell_p)^{\frac{1}{2}p(p+1)-2}$$

$$\cdot \sum_{k=0}^{\infty} \sum_{\kappa} \frac{C_{\kappa}(-\frac{1}{2}\Sigma^{-1})}{k! C_{\kappa}(I_p)} \sum_{t=0}^k \sum_{\tau} b_{\kappa, \tau} \sum_{d=0}^{\infty} \sum_{\delta} \frac{(-(n-p-1)/2)_{\delta}}{d!}$$

$$\cdot \ell_1^{k-t-d} (\ell_1 - \ell_p)^{t+d} \sum_{\beta} g_{\tau, \delta}^{\beta} D_{\beta}^{p-1} \left(\frac{p+2}{2} \right),$$

$$\infty > \ell_1 > \ell_p > 0$$

where $\kappa, \tau, \delta, \beta, b_{\kappa, \tau}, g_{\tau, \delta}^{\beta}$ and D_{β}^{p-1} are exactly as defined in the previous section for (3.3). Further analogous to the case $S_1 S_2^{-1}$ in the special case where $(n-p-1)/2$ is an integer the summation over d becomes finite. In the null case i.e. when $\Sigma = \lambda I_p$ the joint

density of ℓ_1 and ℓ_p was derived by Sugiyama (1970).

Remark 3:

The author has also derived the exact expressions for the joint nonnull density of the largest and smallest roots of (i) MANOVA matrix and (ii) Canonical correlation matrix by using exactly the same technique as in the case of $S_1 S_2^{-1}$. But these expressions are rather long and so are not given here.

REFERENCES

- CONSTANTINE, A.G. (1963). Some noncentral distribution problems in multivariate analysis. *Ann. Math. Statist.*, **34**, 1270-1285.
- JAMES, A.T. (1964). Distributions of matrix variates and latent roots derived from normal samples. *Ann. Math. Statist.*, **35**, 475-501.
- KHATRI, C.G. and PILLAI, K.C.S. (1968). On the noncentral distributions of two test criteria in multivariate analysis of variance. *Ann. Math. Statist.*, **39**, 215-226.
- ROY, S.N. (1957). *Some Aspects of Multivariate Analysis*, John Wiley & Sons, New York.
- SUGIYAMA, T. (1967). On the distribution of the largest latent root of the covariance matrix. *Ann. Math. Statist.*, **38**, 1148-1151.
- SUGIYAMA, T. (1970). Joint distribution of the extreme roots of a covariance matrix. *Ann. Math. Statist.*, **41**, 655-657.
- WAIKAR, V.B. (1971). *Exact joint density of the largest and smallest roots of the Wishart and MANOVA matrices*. Unpublished manuscript.

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